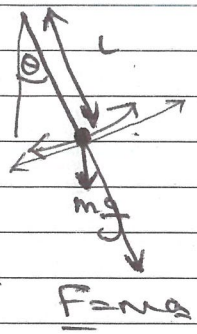
Example

Consider simple harmonic motion (e.g. a pendulum) without friction that is forced at the same frequency as the oscillation

$$\frac{d^2\theta}{dt^2} + \omega^2\theta = \frac{f_0}{l}\cos(\omega t)$$



$F = ma$

Homogeneous problem: $\theta'' + \omega^2\theta = 0$

Soln $\theta = e^{mt}$ $m^2 + \omega^2 = 0$ $\Rightarrow m_{1,2} = \pm i\omega$

$\theta(t) = c_1\cos(\omega t) + c_2\sin(\omega t)$

"complementary soln"

Row 3 of table (p45): $\beta = \omega$ $\alpha = 0$ $k = 0$

$\alpha + i\beta = i\omega$ single resonance $s = 1$

Trial soln $\theta = [B\cos(\omega t) + C\sin(\omega t)]t$

Substitute into nonhomogeneous ODE:

$$\cos(\omega t)\left[2C\omega - tB\omega^2 + \omega^2 tB\right] + \sin(\omega t)\left[-2B\omega - tC\omega^2 + \omega^2 tC\right] = \frac{f_0}{l}\cos(\omega t)$$

Coeff of $\cos(\omega t)$: $2C\omega = f_0/l$

Coeff of $\sin(\omega t)$: $-2B\omega = 0$

Particular soln $\theta_p(t) = \frac{f_0}{2l\omega}\sin(\omega t)\cdot t$

General soln $\theta(t) = \theta_c(t) + \theta_p(t)$

$= c_1\cos(\omega t) + c_2\sin(\omega t) + \frac{f_0 t}{2l\omega}\sin(\omega t)$

"resonance"

Oscillations grow without bound.

Matrix handout

$$A = \begin{pmatrix} 3 & 2-i \\ 4+3i & -5+2i \end{pmatrix} \qquad A^T = \begin{pmatrix} 3 & 4+3i \\ 2-i & -5+2i \end{pmatrix}$$

$$\bar{A} = \begin{pmatrix} 3 & 2+i \\ 4-3i & -5-2i \end{pmatrix}$$

$$\underline{y} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \qquad \underline{x}^T = (x_1\ x_2)$$

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \qquad M_{11} = |a_{22}|$$

Activity A.2

$$\left(\begin{array}{ccc|ccc} 1 & -1 & -1 & 1 & 0 & 0 \\ 3 & -1 & 2 & 0 & 1 & 0 \\ 2 & 2 & 3 & 0 & 0 & 1 \end{array}\right)$$

Use row operations to make LHM the identity

$r_2 \to r_2 - 3r_1$

$$\left(\begin{array}{ccc|ccc} 1 & -1 & -1 & 1 & 0 & 0 \\ 0 & 2 & 5 & -3 & 1 & 0 \\ 2 & 2 & 3 & 0 & 0 & 1 \end{array}\right)$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 7/10 & -1/10 & 3/10 \\ 0 & 1 & 0 & 1/2 & -1/2 & 1/2 \\ 0 & 0 & 1 & -1/5 & 2/5 & -1/5 \end{array}\right)$$

A^{-1}

Check $AA^{-1} = I$

Example A.3 $\qquad \underline{\underline{A}} = \begin{pmatrix} \sin t & t \\ 1 & \cos t \end{pmatrix}$

$$\frac{d\underline{\underline{A}}}{dt} = \begin{pmatrix} \cos t & 1 \\ 0 & -\sin t \end{pmatrix}$$

$$\int_0^{\pi} \underline{\underline{A}}\, dt = \begin{pmatrix} \int_0^{\pi} \sin t\, dt & \int_0^{\pi} t\, dt \\ \int_0^{\pi} 1\, dt & \int_0^{\pi} \cos t\, dt \end{pmatrix} = \begin{pmatrix} [-\cos t]_0^{\pi} & [t^2/2]_0^{\pi} \\ [t]_0^{\pi} & [\sin t]_0^{\pi} \end{pmatrix}$$

$$= \begin{pmatrix} 2 & \pi^2/2 \\ \pi & 0 \end{pmatrix}$$

Linear systems $\qquad \underline{A}\,\underline{x} = \underline{b}$ solve for $\underline{x}$

$$A^T A \underline{x} = A^T \underline{b}$$

$$\underline{x} = A^{-1} \underline{b}.$$

Activity B.3

- If $\underline{x} \neq \underline{0}$ and $c \neq 0$ then $c\underline{x} \neq \underline{0}$
- If $\underline{A}\underline{x} = \lambda \underline{x}$ then

$$\underline{A}(c\underline{x}) = c\underline{A}\underline{x} = c\lambda\underline{x} = \lambda(c\underline{x})$$

($c\underline{x}$: eigenvector; λ: eigenvalue; $c\underline{x}$: eigenvector.)

Activity B.4

$$|\underline{A} - \lambda \underline{I}| = 0 \qquad \underline{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \lambda\underline{I} = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$$

$$\begin{vmatrix} 5-\lambda & -1 \\ 3 & 1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda_1 = 2 \quad \lambda_2 = 4 \quad \text{eigenvalues}$$

Eigenvectors

- $\lambda_1 = 2$ Solve $\underline{A}\underline{x} = 2\underline{x}$

$$\Rightarrow \begin{pmatrix} 5 & -1 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 2 \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$\Rightarrow 5x_1 - x_2 = 2x_1 \Rightarrow x_1 = x_2/3$$
$$3x_1 + x_2 = 2x_2 \Rightarrow x_2 + x_2 = 2x_2$$

$$\Rightarrow \text{eg } \underline{x} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

eigenvector.

- $\lambda_2 = 4 \quad \underline{x}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

Check $\underline{A}\underline{x} = \lambda \underline{x}$

Activity B.6